WORKSHEET 5.8

Sketching a Function from Its Derivatives

Unit 5: Analysing Functions

Key ideas

- Collect, in order: domain and intercepts; asymptotes; where f' is positive or negative; where f'' is positive or negative.
- The signs of f' and f'' together give one of four shapes on each interval: rising or falling, bending up or down.
- A horizontal asymptote comes from a limit as $x \to \pm\infty$; a vertical asymptote from an infinite one-sided limit.
- Mark relative extrema and points of inflection as points, with both coordinates.

Worked examples

Example 1 · A rational function

Describe the graph of $f(x) = \dfrac{x}{x-2}$.

Solution. Vertical asymptote $x = 2$; horizontal asymptote $y = 1$. $f'(x) = -\dfrac{2}{(x-2)^2} < 0$.

Decreasing on each side of $x = 2$; it passes through the origin.

Example 2 · Four shapes

On an interval, $f' > 0$ and $f'' < 0$. Describe the graph there.

Solution. Positive first derivative: rising. Negative second derivative: concave down.

Rising, and levelling off.

Example 3 · Putting it together

List the key features of $f(x) = x^3 - 3x^2$.

Solution. $f'(x) = 3x(x-2)$: maximum $(0, 0)$, minimum $(2, -4)$. $f''(x) = 6x - 6$: inflection $(1, -2)$. Zeros at $x = 0$ and $x = 3$.

Rises to $(0, 0)$, falls through $(1, -2)$ to $(2, -4)$, then rises through $(3, 0)$.

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Practice

CORE

1. Find the vertex and the x-intercepts of $f(x) = x^2 - 4x + 3$.

2. Write equations for the asymptotes of $f(x) = \dfrac{2x+1}{x-3}$.

3. On an interval, $f' < 0$ and $f'' > 0$. Describe the graph there.

4. Describe the graph of $f(x) = e^{-x}$: direction, concavity and asymptote.

EXAM LEVEL

5. Find the relative extrema and the x-coordinates of the points of inflection of $f(x) = x^4 - 4x^2$.

6. Find the maximum, the points of inflection and the horizontal asymptote of $f(x) = \dfrac{1}{x^2+1}$.

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7. A function has $f(1) = -2$, $f'(x) < 0$ for $x < 1$, $f'(x) > 0$ for $x > 1$, and $f''(x) > 0$ for all x. Describe its graph.

8. Find the relative extremum, the point of inflection and the horizontal asymptote of $f(x) = (x+1)e^x$.

Question 9 · Free-response practice

Let $f(x) = \dfrac{x^2 - 1}{x^2 + 1}$.

(a) Find the horizontal asymptote and the intercepts of the graph of f.

(b) Find $f'(x)$ and the coordinates of the relative extremum of f. Classify it.

(c) Given that $f''(x) = \dfrac{4\left(1 - 3x^2\right)}{\left(x^2 + 1\right)^3}$, find the x-coordinates of the points of inflection.

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Worked solutions

1. Vertex $(2, -1)$; intercepts $x = 1$ and $x = 3$.

2. $x = 3$ and $y = 2$.

3. Falling, and concave up (levelling off).

4. Decreasing, concave up, with horizontal asymptote $y = 0$ as $x \to \infty$.

5. $f'(x) = 4x\left(x^2 - 2\right)$: maximum $(0, 0)$, minima $\left(\pm\sqrt{2}, -4\right)$. $f''(x) = 12x^2 - 8$: inflection at $x = \pm\sqrt{\frac{2}{3}}$.

6. $f'(x) = -\dfrac{2x}{(x^2+1)^2}$: maximum $(0, 1)$. $f''(x) = \dfrac{6x^2 - 2}{(x^2+1)^3}$: inflection at $x = \pm\dfrac{1}{\sqrt{3}}$. Asymptote $y = 0$.

7. It falls to a minimum at $(1, -2)$ and then rises, concave up throughout: a single bowl shape.

8. $f'(x) = (x + 2)e^x$: minimum $(-2, -e^{-2})$. $f''(x) = (x + 3)e^x$: inflection at $x = -3$. Asymptote $y = 0$ as $x \to -\infty$.

9. (a) $y = 1$. Intercepts: $(0, -1)$, $(1, 0)$ and $(-1, 0)$.

   (b) $f'(x) = \dfrac{4x}{(x^2+1)^2}$ changes from negative to positive at $x = 0$: a minimum at $(0, -1)$.

   (c) f'' changes sign where $1 - 3x^2 = 0$: $x = \pm\dfrac{1}{\sqrt{3}}$.

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